

USE CASE 5 : AI-Generated Testing & Edge Case Simulation

Covers:

- AI in Testing
 - Test generation
 - Edge cases
 - Output validation
 - Retry vs re-prompt
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Scenario:

The Cart and Payment modules need robust testing.

The AI Must:

- Generate unit tests
 - Generate boundary tests
 - Simulate:
 - Out-of-stock
 - Payment timeout
 - Invalid coupon
 - Produce structured test cases
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Example Prompt:

"Generate JUnit test cases for CartService.

Cover:

- Normal flow
- Empty cart
- Large quantity
- Concurrent modification
- Invalid product ID

Output:

- Test name
 - Scenario
 - Expected result
 - Code snippet"
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What to Build:

- AI-generated test suite
- Edge case documentation

- Coverage report
- Retry demonstration for failed schema

Deliverables:

- Test code
 - Coverage proof
 - Edge case document
 - Security scenario test
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How This Case Study Covers All Topics

Topic	Covered In
Prompting	Use Case 1–5
Output Control	2,4,5
LLM Basics	1
AI Requirements	1
Backend	2
Frontend	3
Code Review	4
Testing	5